DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech Branch : Electrical Engineering/Electrical and Electronics Engineering/Electrical and Power/Electronics and Power

Semester : III

Subject Code & Name: BTEEC302 Electrical Machines-I

Duration: 3 Hr.

Max Marks: 60 **Date:07/02/2025**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Select the correct alternative (Compulsory Question)**					**12**
1	The constant losses in transformer is/are ----				Remember	1
	a. Copper loss	b. Eddy current loss	c. hysteresis loss	d. both b & c		
2	In a three phase star-delta transformer what is the angle difference between primary and secondary phase voltage?				Remember	1
	a. Delta side lags by -30°	b. star side lags by -30°	c. star side leads 30°	d. delta side leads 30°		
3	In dc generator if field winding attains the critical resistance____				Understand	1
	a. machine will generate maximum voltage	b. machine will generate maximum power	c. field winding will burn	d. the voltage generated will be zero		
4	Inter pole winding is connected in -------				Remember	1
	a. series with armature	b. series with main pole	c. parallel with armature	d. parallel with main pole		
5	Cross magnetization effect of armature reaction affects which of the following?				Understand	1
	a. commutation	b. reduction of main field flux	c. reduces the terminal voltage	d. both 2 & 3		
6	For starting a dc motor a starter is required to				Remember	1

	a. limits the starting current	b. [illegible]	c. limits the speed of the motor	d. none of the above		
7	In a 6 pole dc machine: wave winding is used. The number of parallel paths are -----				Apply	1
	a. 6	b. 4	c. 2	d. 1		
8	Maximum power will be developed when back emf is equal to --				Remember	1
	a. supply voltage	b. half of the supply voltage	c. double the supply voltage	d. all of the above		
9	If the field connection of a dc shunt motor is changed then ----				Understand	1
	a. it will run in the same direction by slowly	b. motor will not run	c. it will run in opposite direction	d. it will run in same direction		
10	Transformer core is laminated to reduce -------				Remember	1
	a. eddy current loss	b. hysteresis loss	c. both A & B	d. none of above		
11	Scott connections are used for ----				Apply	1
	a. Single phase to three phase connection	b. Three phase to single phases connection	c. three phase to two phases	d. any of above		
12	If a hybrid stepper motor has a rotor pitch of 36^0 and a step angle of 9^0 the number of its phases must be ---				Evaluate	1
	a. 4	b. 2	c. 3	d. 6		
Q. 2	**Solve the following.**					12
A)	Justify "Transformer is a constant flux device."				Remember	6
B)	A 40 KVA transformer has iron loss of 450 Watt and full load copper loss of 850 watt, If the power factor of the load is 0.8 lagging, Find:- i) Full load efficiency ii) The load at which maximum efficiency occurs iii) The maximum efficiency				Evaluate	6
Q.3	Solve the following.					12

A)	Explain the construction of three phase core type transformer	Understand	6
B)	A 3 phase delta – star step down transformer delivers power to a balanced 3-phase load of 120 KVA at 0.8 power factor. The input line voltage is 11kV and the transformation ratio of the transformer is 0.10. Determine the line voltages, line currents, phase voltages and phase currents on both sides of a transformer.	Evaluate	6
Q. 4	**Solve Any Two of the following.**		**12**
A)	Derive the emf equation of Generator. How it will be implemented to lap and wave connected armature winding?	Apply	**6**
B)	A DC generator has an armature emf of 100 Volts when the useful flux per pole is 20 mWb and speed is 800 rpm. Calculate the generated emf. i) With the same flux and a speed of 1000 rpm. & ii) With the flux per pole 24 mWb and speed of 900rpm	Evaluate	**6**
C)	Explain armature reaction? What are its effects?	Understand	**6**
Q.5	Solve Any Two of the following.		12
A)	Derive the relation between armature current and armature torque for dc motor. Hence obtain this equation for dc series and shunt motor.	Understand	6
B)	A 250 volt dc shunt motor on no load runs at 1000 rpm and takes 5 amp. from supply. Its armature and shunt field resistances are 0.2Ω and 250 Ω respectively. Calculate the speed when loaded taking a current of 50 amp. The armature reaction weakens the field by 3%.	Evaluate	6
C)	Illustrate the principle of DC motor. What is the significance of back emf?	Apply	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the construction and working principle of BLDC motor with its applications.	Understand	6
B)	Write a short note on Stepper Motor.	Understand	6
C)	Explain the different types of magnetic system	Understand	6

*** End ***